JAPANESE
INDUSTRIAL
STANDARD

Translated and Published by
Japanese Standards Association

JIS K 0099 : 2004

(JEMCA/JSA)

Methods for determination of ammonia in flue gas

ICS 13.040.40; 71.040.40

Reference number : JIS K 0099 : 2004 (E)

Foreword

This translation has been made based on the original Japanese Industrial Standard revised by the Minister of Economy, Trade and Industry through deliberations at the Japanese Industrial Standards Committee, as the result of proposal for revision of Japanese Industrial Standard submitted by Japan Environmental Measurement and Chemical Analysis Association (JEMCA)/Japanese Standards Association (JSA) with the draft being attached, based on the provision of Article 12 Clause 1 of the Industrial Standardization Law applicable to the case of revision by the provision of Article 14. Consequently **JIS K 0099** : 1998 is replaced with this Standard.

In this time, the format has been looked over and the revision has been carried out.

Attention is drawn to the possibility that some parts of this Standard may conflict with a patent right, application for a patent after opening to the public, utility model right or application for registration of utility model after opening to the public which have technical properties. The relevant Minister and the Japanese Industrial Standards Committee are not responsible for identifying the patent right, application for a patent after opening to the public, utility model right or application for registration of utility model after opening to the public which have the said technical properties.

Date of Establishment: 1969-05-01

Date of Revision: 2004-01-20

Date of Public Notice in Official Gazette: 2004-01-20

Investigated by: Japanese Industrial Standards Committee

Standards Board

Technical Committee on Environment and Recycling Policy

JIS K 0099:2004, Second English edition published in 2005-01

Translated and published by: Japanese Standards Association
4-1-24, Akasaka, Minato-ku, Tokyo, 107-8440 JAPAN

In the event of any doubts arising as to the contents, the original JIS is to be the final authority.

Contents

Methods for determination of ammonia in flue gas

1 Scope This Japanese Industrial Standard specifies the methods for determination of ammonia in flue gas.

Remarks : In this Standard, flue gas means ones which are generated by being accompanied with combustion, chemical reaction, denitration process, metal surface treatment process, etc., and exhausted out to flue, chimney, or duct, etc. (hereafter referred to as "duct").

2 Normative references Standards listed in Attached Table 1 contain provisions which, through reference in this Standard, constitute provisions of this Standard. The most recent editions of the standards (including amendments) shall be applied.

3 Common matters The common matters concerned with chemical analysis, sampling method of flue gas, molecular absorption spectrophotometry and ion chromatography shall be in accordance with **JIS K 0050**, **JIS K 0095**, **JIS K 0115** and **JIS K 0127** respectively.

4 Classification of analytical methods and their outlines Classification of analytical methods and their outlines shall be in accordance with Table 1.

Table 1 Classification of analytical methods and their outlines

Classification of analytical methods	Outlines of analytical method			Applicable condition
	Summary	Sampling	Range of determination volppm (mg/m^3)	
Indophenol blue absorption spectrophotometry	After absorbing ammonia in flue gas into boric acid solution, add sodium phenolpentacyanonitrosylferrate (III) solution and sodium hypochlorite solution. Indophenol blue is generated, measure its absorbance (640 nm).	Absorption bottle method Absorption solution: boric acid solution (5 g/L) Solution volume: 50 ml × 2 bottles Standard sampling volume: 20 L	1.6 to 15.5 (1.2 to 11.8)	according to **7.1.1**.
Ion chromatography	After absorbing ammonia in flue gas into boric acid solution, introduce it into the ion chromatograph and obtain the chromatogram of ammonium ion.	Absorption bottle method Absorption solution: boric acid solution (5 g/L) Solution volume: 25 ml (1) × 2 bottles or 50 ml (2) × 2 bottles Standard sampling volume: 20 L	0.6 to 15.5 (3) (0.5 to 11.8) 6.2 to 310 (4) (4.7 to 236)	

Notes (1) Absorption solution volume when an absorption bottle of 100 ml in capacity is used.

(2) Absorption solution volume when an absorption bottle of 250 ml in capacity is used.

(3) The case where 50 ml of the absorption solution passed through of sample gas is diluted to 100 ml and made as the sample solution for analysis. The ranges of determination indicated herein are obtained from the standard sampling volume of sample gas, the volume of sample solution for analysis, the reference solution of ammonium ion (0.01 mg/ml) and the optimum range of working curve.

(4) The case where 100 ml of the absorption solution passed through of sample gas is diluted to 250 ml and made as the sample solution for analysis. The ranges of determination indicated herein are obtained from the standard sampling volume of sample gas, the volume of sample solution for analysis, the reference solution of ammonium ion (0.1 mg/ml) and the optimum range of working curve. When the concentration exceeding the range of determination is measured, the sample solution for analysis shall be diluted so as to be included within the range of determination, and measured.

Remarks 1 The units of volppm and mg/m^3 indicated in this Standard are the volume concentration and mass concentration under the standard condition (0 °C, 101.32 kPa).

2 There is the detecting tube method as the simplified analytical method for ammonia in flue gas. The sample gas is sucked into a detecting tube, reacted with the reagent in the tube and the concentration of ammonia is obtained from the changed colour length of reagent. Although the determination range of concentration depends on the classification of detecting tube and the suction volume of sample gas, that agrees within 1 volppm to 1 000 volppm by a suction of 100 ml.

3 In addition to the methods as shown in Table 1, this Standard specifies Annex 1 (normative) Ion electrode method and Annex 2 (normative) Analytical method for the case of interference by coexisting sulfur oxides, etc. in the indophenol blue absorption spectrophotometry.

5 Sampling method of gas

5.1 Sampling of gas For the sampling position of gas to be used for analysis, the points where the representative gas is collected shall be selected, and the sample gas shall be usually taken not less than twice at the same taking position within close times and used for each analysis.

5.2 Reagents and preparation of absorption solution The reagents and preparation of absorption solution shall be as follows:

a) **Water**, class A2 specified in **JIS K 0557**.

b) **Boric acid**, specified in **JIS K 8863**.

c) **Absorption solution [boric acid solution (5 g/L)]** Dissolve 5.0 g of boric acid in water and make 1 L in whole volume.

5.3 Apparatus The apparatus shall be as follows:

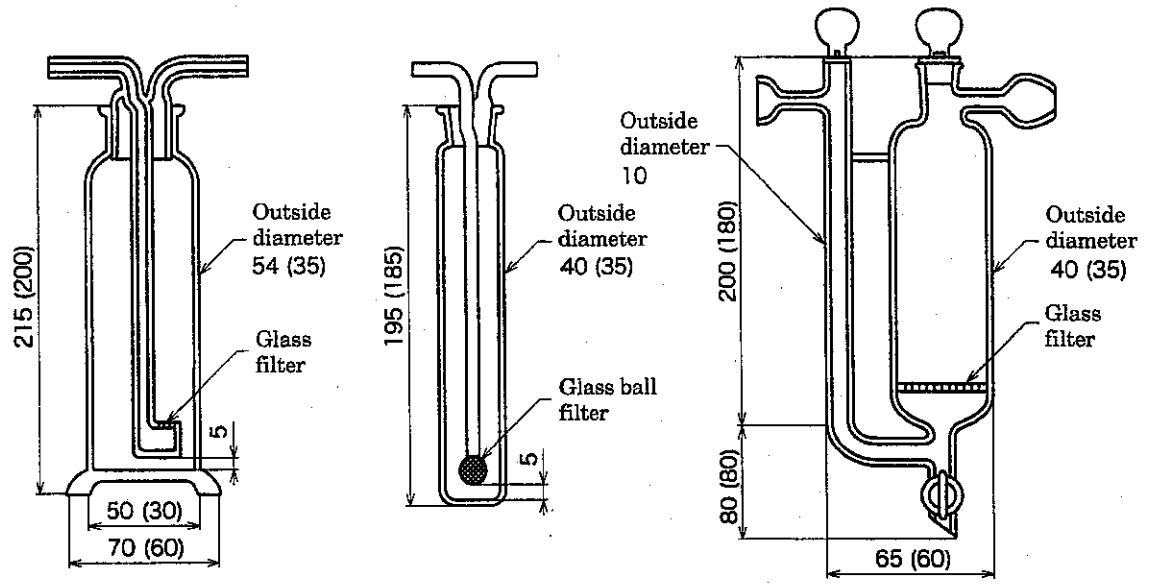
a) **Absorption bottle** The absorption bottle shall be as follows:

1) **Case of indophenol blue absorption spectrophotometry** The two connected absorption bottles (250 ml in capacity) as shown in Fig. 1 shall be used.

2) **Case of ion chromatography** The two connected absorption bottles (100 ml or 250 ml in capacity) as shown in Fig. 1 shall be used.

Unit: mm

The mark of ( ) in the figure shall indicate 100 ml use.



Glass filter and glass ball filter: G1 or G2

a) **Glass filter type** b) **Glass ball type** c) **Glass filter type with branched tube**

Fig. 1 An example of absorption bottle (100 ml, 250 ml)

b) **Gas sampling apparatus** The gas sampling apparatus shall be constructed as shown in Fig. 2 and provided with the following requirements.

1) For the gas sampling tube (B), a glass tube, quartz tube, stainless steel tube (5) or polytetrafluoroethylene tube, etc., which is not corroded by ammonia and coexisting components in the flue gas or not adsorb them, shall be used.

2) For prevention of the dust from mixing with the sample gas, filter media (6) shall be stuffed into a pointed end or suitable portion of the gas sampling tube.

3) For prevention of moisture in sample gas from condensing, the portion between the sampling tube and the cock (P_1) shall be constructed to heat (7).

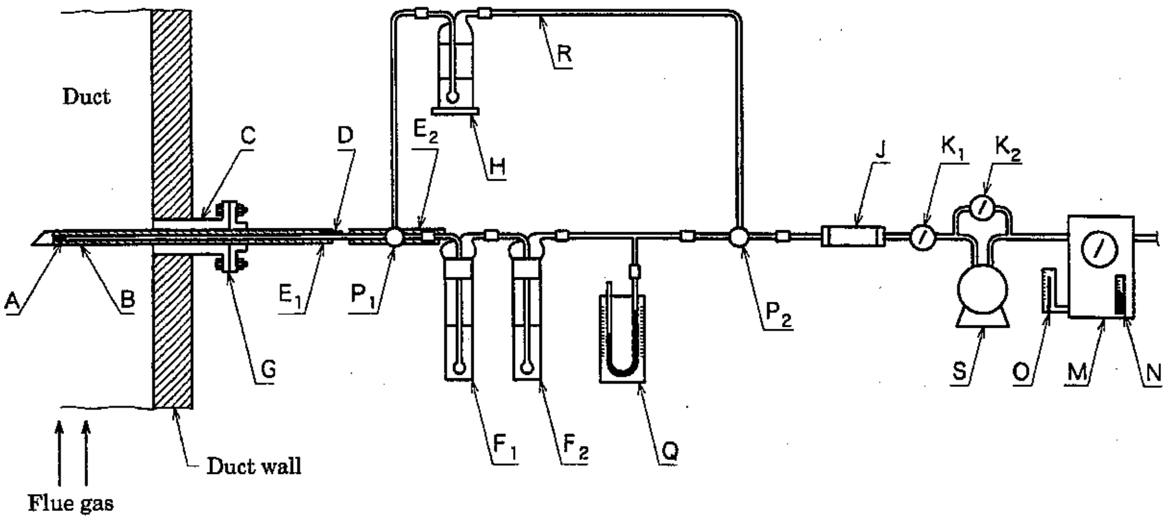
For the connection of tubes at this portion, a ground glass joint, silicone rubber tube or polytetrafluoroethylene tube, etc. shall be used.

4) Each connecting part of the apparatus shall be assembled so as not to generate the gas leakage.

Notes (5) SUS 316 and SUS 316 L will give negative error at the measurement of ammonia in the case of existence of nitrogen oxides.

(6) The quality of material, which does not react chemically with the components in flue gas, for example silica wool, non-alkali glass wool, shall be used.

(7) The piping shall be short as much as possible. When there is the possibility that moisture will be condensed, the portion between the sampling tube and the cock (P_1) shall be heated approximately at 120 °C.



A : Filter media
B : Gas sampling tube
C : Sampling hole
D : Thermometer
E_1, E_2 : Heater
F_1, F_2 : Absorption bottle (250 ml or 100 ml in capacity)
G : Flange
H : Gas washing bottle [contained with 50 ml of sodium hydroxide solution (40 g/L)]
J : Drying tube
K_1, K_2 : Flow rate adjusting cock
M : Wet type or dry type gas meter
N : Thermometer
O : Manometer
P_1, P_2 : Three way stop cock for switching of flow passage
Q : Mercury manometer
R : Bypass
S : Suction pump

Fig. 2 An example of gas sampling apparatus

5.4 Sampling operation The operation shall be as follows. The symbols of the apparatus as shown herein shall be in accordance with Fig. 2.

a) The absorption bottle and absorption solution volume shall be in accordance with either of the following.

1) **Case of indophenol blue absorption spectrophotometry** Pour 50 ml of absorption solution prepared in **5.2 c)** into each absorption bottle (250 ml in capacity) (F_1, F_2) respectively.

2) **Case of ion chromatography** Pour 25 ml of absorption solution prepared in **5.2 c)** into each absorption bottle (100 ml in capacity) (F_1, F_2) respectively.

When the concentration of ammonia is expected to exceed 16 volppm, pour 50 ml of absorption solution prepared in **5.2 c)** into each absorption bottle (250 ml in capacity) (F_1, F_2) respectively.

b) After turning the three way switching stop cock (P_1, P_2) to the side of bypass (R), adjust the flow rate at 1 L/min to 2 L/min previously, and replace the inside of the portion from gas sampling tube (B) to cock (P_1) with the sample gas by operating suction pump (S).

Remarks 1 When the distance from gas sampling tube (B) to absorption bottle is short, no leakage is found in the sampling apparatus and the flow rate can be constantly controlled, it is not required to equip the bypass (R) between P_1 and P_2 in Fig. 2.

2 When no leakage in the sampling apparatus can be confirmed by other means, it is not required to equip the mercury manometer (Q) in Fig. 2.

c) After stopping the suction pump (S), turn the three way switching stop cock (P_1, P_2) to the side of absorption bottles (F_1, F_2). Next, read out the indication (V_1) of gas meter (M) to the nearest 0.01 L.

d) Start the suction pump (S) and pass the sample gas into absorption bottles (F_1, F_2)(8). Adjust the flow rate to approximately 1 L/min to 2 L/min by regulating the flow rate adjusting cocks (K_1, K_2).

Note (8) When the temperature of sampling gas is comparatively high and the absorption solution has a chance to get warm, the absorption bottles are kept in a cooling vessel. In this case, it is recommended to use the absorption bottle as shown in Fig. 1 **a**).

e) Measure and record the temperature (t) and gauge pressure (P_m) of gas meter (M) and the atmospheric pressure (Pa).

f) After taking approximately 20 L(9) of the sample gas, stop the suction pump (S), close the three way switching stop cocks (P_1, P_2) and read out the indication (V_2) of gas meter (M) to the nearest 0.01 L.

Note (9) That may be increased or decreased depending on the concentration of ammonia.

g) As occasion demands, measure the moisture in the sample gas according to clause **6** of **JIS Z 8808**.

5.5 Sampling volume of gas Calculate the sampling amount of gas under the standard condition (0 °C, 101.32 kPa) as dry gas amount (V_{SD}) or wet gas amount (V_{SW}) according to the following formula.

a) **Calculation for obtaining the volume of dry gas**

1) **When the wet type gas meter is used**

$$V_{SD} = V \times \frac{273.15}{273.15+t} \times \frac{P_a + P_m - P_v}{101.32} + 22.41(a+b)$$

2) **When the dry type gas meter is used**

$$V_{SD} = V \times \frac{273.15}{273.15+t} \times \frac{P_a + P_m}{101.32} + 22.41(a+b)$$

b) **Calculation for obtaining the volume of wet gas**

1) **When the wet type gas meter is used**

$$V_{SW} = V \times \frac{273.15}{273.15+t} \times \frac{P_a + P_m - P_v}{101.32} + 22.41(a+b+c)$$

2) **When the dry type gas meter is used**

$$V_{SW} = V \times \frac{273.15}{273.15+t} \times \frac{P_a + P_m}{101.32} + 22.41(a+b+c)$$

where, V_{SD} : volume of dry gas (L)

V_{SW} : volume of wet gas (L)

V : gas volume measured with gas meter (L)
$V_2 - V_1$ at the operation in **5.4 c)** to **f)**

t : temperature indicated on the gas meter (°C)

P_a : atmospheric pressure (kPa)

P_m : gauge pressure indicated on the gas meter (kPa)(10)

P_v : saturated steam pressure at t °C (kPa)

a (10) : target gas for analysis which is collected in the absorption bottle (mol)

b (10) : other gas than the target gas for analysis, which is collected in the absorption bottle (mol)

c (10) : moisture content obtained according to clause **6** of **JIS Z 8808** (mol)

22.41 : gas volume of 1 mol under the standard condition (L)

Note (10) There are many cases that neglect may be permitted.

6 Preparation of sample solution for analysis

6.1 For indophenol blue absorption spectrophotometry The preparation of sample solution for analysis shall be as follows:

a) After finishing the operation of **5.4**, transfer the solution contained in the absorption bottles (F_1, F_2) into a 300 ml beaker and wash the absorption bottles, etc. with absorption solution [boric acid solution (5 g/L)].

b) Transfer the solution contained in the beaker, with washing by absorption solution, into a 250 ml volumetric flask and add further absorption solution up to the marked line. Take this solution as the sample solution for analysis.

6.2 Case of ion chromatography The preparation of sample solution for analysis shall be as follows:

a) After finishing the operation of **5.4**, transfer the solution contained in the absorption bottles (F_1, F_2) into a 300 ml beaker and wash the absorption bottles, etc. with water.

b) When an absorption bottle of 100 ml in capacity is used, transfer the solution contained in the beaker, with washing by using water, into a 100 ml volumetric flask and add water up to the marked line. Take this solution as the sample solution for analysis (11).

Note (11) When solid matter is found in the sample solution for analysis, that shall be previously removed by filtering with using the filter media with pore size of not more than 0.45 μm for preventing the stoppage of separation column.

c) When an absorption bottle of 250 ml in capacity is used, transfer the solution contained in the beaker, with washing by using water, into a 250 ml volumetric flask and add water up to the marked line. Take this solution as the sample solution for analysis (11).

7 Method of determination

7.1 Indophenol blue absorption spectrophotometry

7.1.1 Application requirements This method will be affected by some coexisting cases such as at least 100 times of nitrogen dioxide, at least several tens times of amines, at least 10 times of sulfur dioxide and at least equivalent volume of hydrogen sulfide, compared with concentration of ammonia. Because of this reason, this method shall be applicable for the only case that the influence is ignored or removed.

7.1.2 Reagents and preparation of reagent solution The reagents and preparation of reagent solution shall be as follows:

a) **Reagents** The reagents shall be as follows:

1) **Water**, class A2 specified in **JIS K 0557**.

2) **Sodium hydroxide**, specified in **JIS K 8576**.

3) **Sodium hypochlorite solution**, 5 % to 12 % of available chlorine shall be contained.

4) **Ammonium chloride**, specified in **JIS K 8116**.

5) **Ammonium sulfate**, specified in **JIS K 8960**.

6) **Phenol**, specified in **JIS K 8798**.

7) **Sodium pentacyanonitrosylferrate (III) dihydrate**, specified in **JIS K 8722**.

8) **Reference solution ammonium ion (1 000 mgNH_4^+/L)**, class NH_4^+1 000 specified in **JIS K 0034**.

9) **Boric acid**, specified in **JIS K 8863**.

10) **Starch (soluble)**, specified in **JIS K 8659**.

11) **Acetic acid**, specified in **JIS K 8355**.

12) **Potassium iodide**, specified in **JIS K 8913**.

13) **Potassium iodate**, specified in **JIS K 8005**.

14) **Magnesium perchlorate**, specified in **JIS K 8228**.

b) **Preparation of reagent solution** The preparation of reagent solution shall be as follows:

1) **Sodium hypochlorite solution** Dissolve 600/N ml (12) of sodium hypochlorite solution (30 g/L to 100 g/L of available chlorine), 15 g of sodium hydroxide in water and make the whole volume up to 1 L. Prepare this solution when used.

2) **Phenol-sodium pentacyanonitrosylferrate (III) solution** Dissolve 5 g of phenol, 25 mg of sodium pentacyanonitrosylferrate (III) dihydrate in water and make the whole volume up to 500 ml. This solution shall be stocked in cool and dark place and the solution elapsed not less than one month shall not be used.

3) **Reference stock solution of ammonium ion (0.1 mg/ml)** Dilute the ammonium ion reference solution (1 000 $mgNH_4^+/L$) of **a) 8)** 10 times, or after allowing to stand ammonium chloride for at least 16 h in a desiccator in which magnesium perchlorate (dry use) was previously contained, weigh out 0.296 5 g of it, dissolve in water, transfer the solution into a 1 000 ml volumetric flask with washing and add water up to the marked line. Ammonium sulfate may be used instead of ammonium chloride. In this case, previously heat it at 105 °C ± 2 °C for 2 h, cool it in a desiccator contained with magnesium perchlorate as it is, then, weigh out 0.366 3 g of it, carry out the same operation as described above and make the whole volume to 1 000 ml.

4) **Reference solution of ammonium ion (0.001 mg/ml)** Take accurately 5.0 ml of the reference stock solution of ammonium ion prepared in **3)** into a 500 ml volumetric flask and add the absorption solution up to the marked line.

5) **Absorption solution**, prepared in **5.2 c)**.

Note (12) Available chlorine content (g/L) in sodium hypochlorite solution shall be obtained as follows:

Take v ml (10 ml at normal condition) of sodium hypochlorite solution into a 200 ml volumetric flask and add water up to the marked line. Take an aliquot of 10 ml of this solution into a 300 ml Erlenmeyer flask with ground stopper and add water to make approximately 100 ml. Add 1 g to 2 g of potassium iodide, 6 ml of acetic acid (1+1), stopper the flask with plug, shake gently it and leave it in the dark place for 5 min as it is. Titrate free iodine with 0.05 mol/L sodium thiosulfate solution. After the yellow colour of solution becomes faint, add 1 ml of starch solution* as an indicator, continue to titrate and regard the point where the blue colour of starch iodine disappears as the end point. Separately, carry out the blank test under the same condition and correct the titrated value. Calculate available chlorine content (g/L) according to the following formula.

$$N = a' \times f \times \frac{200}{10} \times \frac{1\,000}{v} \times 0.001\,773$$

where, N : available chlorine content (g/L)

a' : volume of 0.05 mol/L sodium thiosulfate solution required for titration (ml)

f : factor of 0.05 mol/L sodium thiosulfate solution (13)

v : sampling volume of sodium hypochlorite solution (ml)

0.001 773 : mass of chlorine equivalent to 1 ml of 0.05 mol/L sodium thiosulfate solution (g)

Note * Mix 1 g of starch (soluble) with approximately 10 ml of water, next add it into 100 ml of hot water while agitating sufficiently, boil it for approximately 1 min and then cool it as it is. This solution shall be prepared when used.

Note (13) The factor (f) of 0.05 mol/L sodium thiosulfate solution shall be obtained as follows:

Dry potassium iodate at 130 °C for approximately 2 h, allow to cool it in a desiccator contained with magnesium perchlorate (dry use) and weigh out 0.35 g to 0.36 g of it to the nearest 0.1 mg. Dissolve it with water, transfer the solution into a 250 ml volumetric flask by water with washing, and add water up to the marked line. Take an aliquot of 25 ml of this solution into a 300 ml Erlenmeyer flask with ground stopper, make up to 100 ml by adding water, add 1 g to 2 g of potassium iodide, 6 ml of acetic acid (1+1), shake gently the flask with a stopper applied and leave it in dark place for 5 min as it is. Titrate free iodine with sodium thiosulfate solution. After the yellow colour of solution becomes faint, add 1 ml of starch solution* as an indicator, continue to titrate and regard the point where the blue colour of starch iodine disappears as the end point. Separately, carry out the blank test under the same condition and correct the titrated value. Calculate the factor according to the following formula.

Note * That shall be in accordance with Note * in Note (12).

$$f = \frac{m \times 25/250}{a' \times 0.001\,783}$$

where, f : factor of 0.05 mol/L sodium thiosulfate solution

m : sampling mass of potassium iodate (g)

a' : volume of 0.05 mol/L sodium thiosulfate solution required for titration (ml)

0.001 783 : mass of potassium iodate equivalent to 1 ml of 0.05 mol/L sodium thiosulfate solution (g)

7.1.3 Determination operation The determination operation shall be as follows:

a) Take 10 ml of the sample solution for analysis prepared in **6.1** into a 25 ml volumetric flask.

b) Add 5 ml of sodium phenol-pentacyanonitrosylferrate (III) solution, and mix it thoroughly with shaking. Then add 5 ml of sodium hypochlorite solution and add water up to the marked line. Mix it gently with the stopper applied and leave it at 25 °C to 30 °C in temperature of solution for 1 h as it is.

c) Take a portion of this solution in an absorption cell and measure the absorbance at neighborhood of 640 nm in wavelength with using the solution, in which the operations of **a**) and **b**) has been carried out on the absorption solution instead of the sample solution for analysis, as the reference solution.

d) Obtain the quantity of ammonium ion (NH_4^+) from the working curve prepared in **7.1.4**.

7.1.4 Preparation of working curve The preparation of working curve shall be as follows:

a) Take stepwise 1.0 ml to 10.0 ml of ammonium ion reference solution (0.001 mg/ml) into 25 ml several volumetric flasks and add the absorption solution to make the volume of solution 10 ml.

b) Carry out the operations of **7.1.3 b**) and **7.1.3 c**) and prepare the relation curve between the quantity of ammonium ion and the absorbance.

Remarks : Care shall be taken on handling because this waste water contains hazardous material.

7.1.5 Calculation The concentration of ammonia in the sample gas shall be calculated according to the following formula.

$$C_{\mathrm{v}} = \frac{1.24 \times (a-b) \times 250/10}{V_{\mathrm{s}}} \times 1\,000$$

$$C_{\mathrm{w}} = \frac{0.944 \times (a-b) \times 250/10}{V_{\mathrm{s}}} \times 1\,000$$

$$C_{\mathrm{w}} = C_{\mathrm{v}} \times 0.760$$

where, C_{v} : volume concentration of ammonia in sample gas (volppm)

C_{w} : mass concentration of ammonia in sample gas (mg/m³)

a : mass of ammonium ion obtained in **7.1.3 d**) (mg)

b : mass of ammonium ion obtained in the blank test of **7.1.3 e)** (mg)

V_s : sampling volume of gas under standard condition calculated according to **5.5** (L)
(V_{SD} is used for the volume of dry gas and V_{SW} is used for the volume of wet gas)

1.24 : volume of ammonia (NH_3) equivalent to 1 mg of NH_4^+ (ml) (standard condition)

0.944 : mass of ammonia (NH_3) equivalent to 1 mg of NH_4^+ (mg) (standard condition)

0.760 : mass concentration of ammonia (NH_3) equivalent to 1 volppm of ammonia (mg/m^3)

7.2 Ion chromatography

7.2.1 Reagents and preparation of reagent solution The reagents and preparation of reagent solution shall be as follows:

a) **Reagents** The reagents shall be as follows:

1) **Water**, class A2 specified in **JIS K 0557**.

2) **Nitric acid**, specified in **JIS K 8541**.

3) **Reference solution of ammonium ion (1 000 mgNH_4^+/L)**, class NH_4^+1 000 specified in **JIS K 0034**.

4) **Ammonium chloride**, specified in **JIS K 8116**.

5) **Oxalic acid dihydrate**, specified in **JIS K 8519**.

6) **Methane sulfonic acid**

b) **Preparation of reagent solution** The preparation of reagent solution shall be as follows:

1) **Eluant** Eluant is varied depending on the kind of apparatus and the kind of separation columns used(14). For the apparatus and separation column to be used, the optimum ones shall be used. The typical example is as follows:

1.1) **Case of using the apparatus with suppressor** That shall be as follows:

1.1.1) **Methane sulfonic acid solution (15 mmol/L)** Dissolve 1 ml of methane sulfonic acid in water, transfer it into a 1 000 ml volumetric flask with washing by water and add water up to the marked line.

1.2) **Case of using the apparatus without suppressor** That shall be as follows:

1.2.1) **Nitric acid solution (1.6 mmol/L to 5.0 mmol/L)** Dissolve a definite volume of 16 ml to 50 ml of nitric acid solution (0.1 mol/L) (dissolve 7.2 ml of nitric acid in water and make the whole volume 1 000 ml), transfer it into a 1 000 ml volumetric flask by washing with water and add water up to the marked line.

1.2.2) **Oxalic acid solution (3.5 mmol/L)** Weigh out 442 mg of oxalic acid dihydrate, dissolve it with a small amount of water, transfer it into a 1 000 ml volumetric flask, and add water up to the marked line. Filter it with membrane filter with 0.45 μm of pore diameter.

2) **Regeneration solution (removal solution)** The regeneration solution shall be the liquid to be used for regenerating or continuous maintaining the function of suppressor and used when the regeneration is carried out electrically or chemically. The optimum one for the kind of apparatus and suppressor shall be used. The typical example is as follows:

2.1) **Case of electrical regeneration** That shall be as follows:

2.1.1) **Water**, class A2 specified in **JIS K 0557.**

2.1.2) **Eluant** The eluant passing through the detector shall be the regeneration solution for the electrodialysis type suppressor.

2.2) **Case of chemical regeneration** That shall be as follows:

2.2.1) **Tetramethyl ammonium hydroxide solution (40 mmol/L)** Dissolve 24.2 ml of tetramethyl ammonium hydroxide solution (15 %) in water and make the whole volume 1 000 ml(15).

3) **Reference stock solution of ammonium ion (1 mg/ml)** That shall be the reference solution of ammonium ion (1 000 mgNH_4^+/L) in **a) 3**). Or, leave ammonium chloride in a desiccator contained with magnesium perchlorate (dry use) for not less than 16 h as it is, weigh out 2.965 g of it, dissolve it with a small amount of water, transfer it into a 1 000 ml volumetric flask and add water up to the marked line.

4) **Reference solution of ammonium ion (0.1 mg/ml)** Take accurately 10 ml of the reference stock solution of ammonium ion (1 mg/ml) prepared in **3**) into a 100 ml volumetric flask and add water up to the marked line. Prepare when used.

5) **Reference solution of ammonium ion (0.01 mg/ml)** Take accurately 10 ml of the reference solution of ammonium ion (0.1 mg/ml) prepared in **4**) into a 100 ml volumetric flask and add water up to the marked line. Prepare when used.

Notes (14) It is recommended to select as referring to the instruction manuals of apparatus and column. After confirming that ammonium ion can be measured quantitatively, other eluant than such one as indicated herein may be used according to the characteristics of separation column.

(15) It is allowed to use other regeneration solution than such one as indicated herein according to the characteristics of separation columns and the kind of eluant.

7.2.2 Ion chromatograph The ion chromatograph shall be as follows:

a) **Sample injection device** Automatic device in which a definite volume of the sample solution is introduced in the apparatus with good reproducibility, or manual operation device in which the sample solution is introduced into the sample injection loop (a definite volume between 10 μl to 250 μl) built in the apparatus with a syringe of 1 ml to 10 ml.

b) **Separation column** Column made of inert synthetic resin or metal which has 2 mm to 8 mm in inside diameter and 30 mm to 300 mm in length and packed with a cation exchanger. The column shall be able to separate the target ion for analysis from the adjacent ion thoroughly.

c) **Pre-column** Column used for concentration, preliminary separation, removal of foreign matter and equipped in front of the separation column, as occasion demands. That is a column made of inert synthetic resin or metal, and of 2 mm to 6 mm in inside diameter and 5 mm to 50 mm in length and packed with a cation exchanger which is the same type as the separation column.

d) **Suppressor**(16) Device in which electric conductivity is reduced by changing eluant electrically or chemically for the purpose of measuring the kind of ion in eluant by an electric conductivity detector with high sensitivity. There are membrane dialysis type, column removal type and suspension resin adsorption type for the suppressor.

e) **Detector**, electric conductivity detector.

Note (16) In the ion chromatograph on the market, there are one type of the apparatus equipped with separation column and suppressor, and another type of that equipped with only single separation column without the suppressor. Both types of apparatus may be used.

7.2.3 Determination operation The operation shall be as follows:

a) Hold the ion chromomatograph ready to measure and keep the eluant flow in the separation column at a definite flow rate (for example, 0.5 ml/min to 2 ml/min) beforehand. For the apparatus equipped with the suppressor, flow the eluant into the inside of the separation column and the suppressor, and further keep the regeneration solution flow at the outside of the suppressor at a definite flow rate.

b) Introduce a definite volume (10 μl to 250 μl)(17) of the sample solution for analysis prepared in **6.2** into the ion chromatograph with a sample injection device and record the chromatogram.

c) Obtain peak area or peak height about the peak corresponding to ammonium ion on the chromatogram.

d) Obtain the concentration of ammonium ion (mg/ml) in the sample solution for analysis from the working curve prepared in **7.2.4**.

e) Take 40 ml of the absorption solution prepared in **5.2 c**) into a 100 ml volumetric flask and add water up to the marked line, then take the same volume of this solution as the injected volume in **b**), carry out the operations corresponding to **a**) to **c**) and obtain the blank test value of ammonium ion(18).

Notes (17) When the sample with especially lower concentration is measured, it is recommended to use the condensed column packed with the same kind of cation exchanger as separation column instead of the sample injection loop (a definite volume between 10 μl and 250 μl) built in apparatus.

(18) When an absorption bottle of 250 ml in capacity is used, take 100 ml of absorption solution into a 250 ml volumetric flask, add water up to the marked line and then carry out the same operation.

7.2.4 Preparation of working curve The preparation of working curve shall be as follows:

a) Take stepwise 1.0 ml to 25.0 ml (19) of the reference solution of ammonium ion (0.01 mg/ml) into 100 ml several volumetric flasks, add water up to the marked line and obtain the concentration of each solution (mg/ml) (20).

b) Carry out the operations according to **7.2.3 a)** to **c)** and obtain the peak area or peak height corresponding to each ammonium ion.

c) Separately, carry out the operations according to **7.2.3 a)** to **c)** with using water as a blank test and obtain the peak area or peak height corresponding to each ammonium ion.

d) Prepare the relation curve between the peak area or peak height and the concentration of ammonium ion (mg/ml) according to the correction with blank test value. The working curve shall be prepared at every measuring time of sample.

Notes (19) According to the expected concentration of the sample, several points are taken in the range between 1.0 ml to 25.0 ml in either 0.1 mg/ml or 0.01 mg/ml of the reference solution of ammonium ion.

(20) When an absorption bottle of 250 ml in capacity is used, 1.0 ml to 50.0 ml of the reference solutions of ammonium ion (0.1 mg/ml) are taken stepwise into 250 ml several volumetric flasks, after adding water up to the marked line, the same operation is carried out and the working curve is prepared.

7.2.5 Calculation The concentration of ammonia in the sample gas shall be calculated according to the following formula.

$$C_v = \frac{1.24 \times (a-b) \times v}{V_s} \times 1\,000$$

$$C_w = \frac{0.944 \times (a-b) \times v}{V_s} \times 1\,000$$

$$C_w = C_v \times 0.760$$

where, C_v : volume concentration of ammonia in sample gas (volppm)

C_w : mass concentration of ammonia in sample gas (mg/m^3)

a : concentration of ammonium ion in the sample solution for analysis obtained in **7.2.3 d)** (mg/ml)

b : concentration of ammonium ion obtained in the blank test of **7.2.3 e)** (mg/ml)

v : capacity of volumetric flask (100 is used for 100 ml, and 250 is used for 250 ml) (ml)

V_s : gas sampling volume under the standard condition, calculated according to **5.5** (L)
(V_{SD} is used for dry gas volume, and V_{SW} is used for wet gas volume)

1.24 : volume of ammonia (NH_3) equivalent to 1 mg of NH_4^+ (ml) (the standard condition)

0.944 : mass of ammonia (NH_3) equivalent to 1 mg of NH_4^+ (mg)

0.760 : mass concentration of ammonia (NH_3) equivalent to 1 volppm of ammonia (mg/m^3)

8 Record of analytical result

8.1 Items of record The items recorded as the analytical result shall be as follows:

a) **Analytical value**

b) **Classification of analytical method**

c) **Sampling date of the gas**

d) **Other items to be required**

8.2 Obtaining method of analytical value of flue gas Carry out the analysis twice about the same sample solution for analysis at every sampling time, obtain the mean value and round off to two significant figures.

Attached Table 1 Normative references

JIS K 0034	*Reference materials—Standard solution—Ammonium ion*
JIS K 0050	*General rules for chemical analysis*
JIS K 0095	*Methods for sampling of flue gas*
JIS K 0115	*General rules for molecular absorptiometric analysis*
JIS K 0127	*General rules for ion chromatographic analysis*
JIS K 0557	*Water used for industrial water and wastewater analysis*
JIS K 8005	*Reference materials for volumetric analysis*
JIS K 8116	*Ammonium chloride*
JIS K 8228	*Magnesium perchlorate*
JIS K 8230	*Hydrogen peroxide*
JIS K 8355	*Acetic acid*
JIS K 8519	*Oxalic acid dihydrate*
JIS K 8541	*Nitric acid*
JIS K 8576	*Sodium hydroxide*
JIS K 8659	*Starch, soluble*
JIS K 8722	*Sodium pentacyanonitrosylferrate (III) dihydrate*
JIS K 8798	*Phenol*
JIS K 8863	*Boric acid*
JIS K 8913	*Potassium iodide*
JIS K 8960	*Ammonium sulfate*
JIS Z 8808	*Methods of measuring dust concentration in flue gas*

Attached Table 2 Saturated steam pressure of water

Unit: kPa

Temperature ℃	$\frac{1}{10}$ ℃ .0	.1	.2	.3	.4	.5	.6	.7	.8	.9
0	0.610 5	0.615 0	0.619 5	0.624 1	0.628 6	0.633 3	0.647 9	0.642 6	0.647 3	0.651 9
1	0.656 7	0.661 5	0.666 3	0.671 1	0.675 9	0.680 9	0.685 8	0.690 7	0.695 8	0.700 7
2	0.705 8	0.710 9	0.715 9	0.721 0	0.726 2	0.731 4	0.736 6	0.741 9	0.747 3	0.752 6
3	0.757 9	0.763 3	0.768 7	0.774 2	0.779 7	0.785 1	0.790 7	0.796 3	0.801 9	0.807 7
4	0.813 4	0.819 1	0.824 9	0.830 6	0.836 5	0.842 3	0.848 3	0.854 3	0.860 3	0.866 3
5	0.872 3	0.878 5	0.884 6	0.890 7	0.897 0	0.903 3	0.909 5	0.915 8	0.922 2	0.928 6
6	0.935 0	0.941 5	0.948 1	0.954 6	0.961 1	0.967 8	0.974 5	0.981 3	0.988 1	0.994 9
7	1.002	1.009	1.016	1.022	1.030	1.037	1.044	1.051	1.058	1.065
8	1.073	1.080	1.087	1.095	1.102	1.110	1.117	1.125	1.132	1.140
9	1.148	1.156	1.164	1.171	1.179	1.187	1.195	1.203	1.211	1.219
10	1.228	1.236	1.244	1.253	1.261	1.269	1.278	1.286	1.295	1.304
11	1.312	1.321	1.330	1.338 8	1.347 8	1.356 7	1.365 8	1.374 8	1.383 9	1.399 8
12	1.402 3	1.411 6	1.420 9	1.430 3	1.439 7	1.449 2	1.458 7	1.468 3	1.477 9	1.487 6
13	1.497 3	1.507 2	1.517 1	1.526 9	1.536 9	1.547 1	1.557 2	1.567 3	1.577 6	1.589 7
14	1.598 1	1.608 5	1.619 1	1.629 6	1.640 1	1.650 8	1.661 5	1.672 3	1.683 1	1.694 0
15	1.704 9	1.715 9	1.726 9	1.738 1	1.749 3	1.760 5	1.771 9	1.783 2	1.794 7	1.806 1
16	1.817 7	1.829 3	1.841 0	1.852 9	1.864 8	1.876 6	1.888 6	1.900 6	1.912 8	1.924 9
17	1.937 2	1.949 4	1.961 8	1.974 4	1.986 9	1.999 4	2.012 1	2.024 9	2.037 7	2.050 5
18	2.063 4	2.076 5	2.089 6	2.102 8	2.116 0	2.129 3	2.142 6	2.156 0	2.169 4	2.183 0
19	2.196 8	2.210 6	2.224 5	2.238 3	2.252 3	2.266 3	2.280 5	2.294 7	2.309 0	2.323 4
20	2.337 8	2.352 3	2.366 9	2.381 5	2.396 3	2.411 1	2.426 1	2.441 0	2.456 1	2.471 3
21	2.486 5	2.501 8	2.517 1	2.532 6	2.548 2	2.563 9	2.579 7	2.595 5	2.611 4	2.627 4
22	2.643 4	2.659 5	2.675 8	2.692 2	2.708 6	2.725 1	2.741 8	2.758 4	2.775 1	2.791 9
23	2.808 8	2.825 9	2.843 0	2.860 2	2.877 5	2.895 0	2.912 4	2.930 0	2.947 8	2.965 5
24	2.983 4	3.001 4	3.019 5	3.037 8	3.056 0	3.074 4	3.092 8	3.111 3	3.129 9	3.148 5
25	3.167 2	3.186 0	3.204 9	3.224 0	3.243 0	3.262 5	3.282 0	3.301 6	3.321 3	3.341 1
26	3.360 9	3.380 9	3.400 9	3.421 1	3.441 3	3.461 6	3.482 0	3.502 5	3.523 2	3.544 0
27	3.564 9	3.586 0	3.607 0	3.628 2	3.649 6	3.671 0	3.692 5	3.714 1	3.735 8	3.757 7
28	3.779 6	3.801 6	3.823 7	3.846 0	3.868 3	3.890 9	3.913 5	3.936 3	3.959 3	3.982 3
29	4.005 4	4.028 6	4.051 9	4.075 4	4.099 0	4.122 7	4.146 6	4.170 5	4.194 5	4.218 6
30	4.242 9	4.267 2	4.291 8	4.316 4	4.341 1	4.365 9	4.390 8	4.415 9	4.441 2	4.466 7
31	4.492 3	4.518 0	4.543 9	4.569 8	4.595 8	4.621 9	4.648 2	4.674 5	4.701 1	4.727 9
32	4.754 7	4.781 6	4.808 7	4.835 9	4.863 2	4.890 7	4.918 4	4.934 1	4.974 0	5.002 0
33	5.030 1	5.058 5	5.086 9	5.115 4	5.144 1	5.173 0	5.202 0	5.231 2	5.260 5	5.289 8
34	5.319 3	5.349 0	5.378 8	5.408 8	5.439 0	5.469 3	5.499 7	5.530 2	5.560 9	5.591 8
35	5.622 9	5.654 1	5.685 4	5.716 9	5.748 5	5.780 2	5.812 2	5.844 3	5.876 6	5.908 8
36	5.941 2	5.973 9	6.006 7	6.039 6	6.072 7	6.106 0	6.139 5	6.173 1	6.207 0	6.241 0
37	6.275 1	6.309 3	6.343 7	6.378 3	6.413 1	6.448 0	6.483 1	6.518 3	6.553 7	6.589 3
38	6.625 1	6.660 9	6.696 9	6.733 0	6.769 3	6.805 8	6.842 5	6.879 4	6.916 6	6.954 1
39	6.991 7	7.029 4	7.067 3	7.105 3	7.143 4	7.181 7	7.220 2	7.258 9	7.297 7	7.336 7
40	7.375 9	7.414	7.454	7.494	7.534	7.574	7.614	7.654	7.695	7.737
41	7.778	7.819	7.861	7.902	7.943	7.986	8.029	8.071	8.114	8.157

Attached Table 2 (continued)

Unit: kPa

Temperature	$\frac{1}{10}$ °C									
°C	.0	.1	.2	.3	.4	.5	.6	.7	.8	.9
42	8.199	8.242	8.285	8.329	8.373	8.417	8.461	8.505	8.549	8.594
43	8.639	8.685	8.730	8.775	8.821	8.867	8.914	8.961	9.007	9.054
44	9.101	9.147	9.195	9.243	9.291	9.339	9.387	9.435	9.485	9.534
45	9.583	9.633	9.682	9.731	9.781	9.831	9.882	9.933	9.983	10.03
46	10.09	10.14	10.19	10.24	10.29	10.35	10.40	10.45	10.51	10.56
47	10.61	10.67	10.72	10.78	10.83	10.88	10.94	10.99	11.05	11.10
48	11.16	11.22	11.24	11.33	11.39	11.45	11.50	11.56	11.62	11.68
49	11.47	11.79	11.85	11.91	11.97	12.03	12.09	12.15	12.21	12.27
50	12.33	12.39	12.46	12.52	12.58	12.64	12.70	12.77	12.83	12.89
51	12.96	13.02	13.09	13.15	13.22	13.28	13.347	13.412	13.479	13.544
52	13.611	13.678	13.746	13.812	13.880	13.948	14.016	14.084	14.154	14.223
53	14.292	14.361	14.431	14.500	14.571	14.641	14.712	14.784	14.856	14.928
54	15.000	15.072	15.144	15.217	15.291	15.364	15.439	15.513	15.588	15.663
55	15.737	15.812	15.887	15.963	16.040	16.117	16.195	16.272	16.349	16.427
56	16.505	16.585	16.664	16.743	16.823	16.903	16.983	17.064	17.145	17.227
57	17.308	17.391	17.473	17.556	17.639	17.721	17.805	17.889	17.973	18.059
58	18.143	18.228	18.313	18.400	18.486	18.573	18.660	18.748	18.836	18.924
59	19.012	19.101	19.190	19.280	19.369	19.460	19.550	19.641	19.732	19.824
60	19.916	20.008	20.101	20.194	20.288	20.381	20.476	20.570	20.665	20.760
61	20.856	20.952	21.048	21.144	21.241	21.340	21.438	21.542	21.636	21.734
62	21.834	21.934	22.034	22.134	22.236	22.337	22.438	22.541	22.634	22.746
63	22.849	22.953	23.057	23.162	23.267	23.373	23.478	23.585	23.691	23.798
64	23.906	24.013	24.121	24.230	24.339	24.449	24.558	24.669	24.779	24.891
65	25.003	25.115	25.227	25.339	25.453	25.567	25.682	25.797	25.911	26.054
66	26.143	26.259	26.376	26.494	26.611	26.728	26.847	26.966	27.086	27.206
67	27.326	27.447	27.568	27.690	27.812	27.935	28.058	28.180	28.304	28.428
68	28.554	28.679	28.806	28.932	29.059	29.186	29.314	29.442	29.570	29.699
69	29.828	29.959	30.090	30.220	30.352	30.484	30.617	30.751	30.884	31.017
70	31.16	31.29	31.42	31.56	31.70	31.84	31.97	32.12	32.25	32.38
71	32.52	32.66	32.80	32.94	33.08	33.22	33.37	33.52	33.65	33.80
72	33.94	34.09	34.24	34.38	34.53	34.68	34.82	34.97	35.13	35.28
73	35.42	35.57	35.73	35.88	36.04	36.18	36.34	36.49	36.65	36.80
74	36.96	37.12	37.26	37.42	37.58	37.74	37.90	38.06	38.22	38.38
75	38.54	38.70	38.86	39.04	39.20	39.36	39.53	39.69	39.85	40.02
76	40.18	40.36	40.52	40.69	40.86	41.02	41.20	41.37	41.54	41.72
77	41.88	42.05	42.22	42.40	42.57	42.76	42.93	43.10	43.29	43.46
78	46.64	43.82	44.00	44.18	44.36	44.54	44.73	44.90	45.09	45.28
79	45.46	45.65	45.84	46.02	46.21	46.40	46.58	46.77	46.96	47.16
80	47.34	47.53	47.73	47.92	48.12	48.32	48.50	48.70	48.90	49.10
81	49.29	49.49	49.69	49.89	50.22	50.30	50.64	50.70	50.90	51.12
82	51.32	51.52	51.73	51.93	52.14	52.36	52.56	52.77	52.98	53.20
83	53.41	53.62	53.84	54.05	54.26	54.48	54.70	54.92	55.13	55.36

Attached Table 2 (concluded)

Unit: kPa

Temperature ℃	1/10 ℃ .0	.1	.2	.3	.4	.5	.6	.7	.8	.9
84	55.57	55.78	56.01	56.22	56.45	56.68	56.90	57.13	57.36	57.58
85	57.81	58.04	58.26	58.49	58.73	58.96	59.18	59.42	59.65	59.89
86	60.12	60.34	60.58	60.82	61.06	61.29	61.53	61.77	62.01	62.25
87	62.49	62.73	62.97	63.21	63.46	63.70	63.95	64.19	64.45	64.69
88	64.94	65.19	65.45	65.69	65.94	66.19	66.45	66.70	66.97	67.22
89	67.47	67.73	67.99	68.25	68.51	68.78	69.03	69.30	69.57	69.70
90	70.096	70.362	70.630	70.898	71.167	71.437	71.709	71.981	72.254	72.527
91	72.801	73.075	73.351	73.629	73.907	74.186	74.465	74.746	75.027	75.310
92	75.592	75.876	76.162	76.447	76.734	77.022	77.310	77.599	77.890	78.182
93	78.474	78.767	79.060	79.355	79.651	79.948	80.245	80.544	80.844	81.145
94	81.447	81.749	82.052	82.356	82.661	82.968	83.274	83.582	83.892	84.202
95	84.513	84.825	85.138	85.452	85.766	86.082	86.400	86.717	87.036	87.355
96	87.675	87.997	88.319	88.643	88.967	89.293	89.619	89.947	90.275	90.605
97	90.935	91.266	91.598	91.931	92.266	92.602	92.939	93.276	93.615	93.954
98	94.295	94.636	94.979	95.323	95.667	96.012	96.359	96.707	97.056	97.407
99	97.757	98.109	98.463	98.816	99.172	99.528	99.885	100.24	100.60	100.96
100	101.32	101.69	102.05	102.42	101.78	103.15	103.52	103.89	104.26	104.63
101	105.00	105.37	105.75	106.12	106.50	106.88	107.26	107.64	108.02	108.40

Annex 1 (normative)
Ion electrode method

1 Scope This Annex specifies the analytical method of ammonia according to ion electrode method.

2 Outline of analytical method The outline of analytical method shall be as follows:

Annex 1 Table 1 Outline of analytical method

Outline of analytical method		
Summary	Sampling	Range of determination volppm (mg/m^3)
After absorbing ammonia in flue gas into boric acid solution, add sodium hydroxide solution and measure free ammonia with membrane type electrode.	Absorption bottle method Absorption solution: boric acid solution (5 g/L) Absorption volume: 50 ml × 2 bottles Standard sampling volume: 20 L	1.8 to 1 600 (1.3 to 1 200)

3 Sampling method of gas

3.1 Sampling of gas The sampling of gas shall be in accordance with clause **5** of the text.

3.2 Reagents and preparation of absorption solution The reagents and preparation of absorption solution shall be as follows:

a) **Water**, class A2 specified in **JIS K 0557**.

b) **Boric acid**, specified in **JIS K 8863**.

c) **Absorption solution [boric acid solution (5 g/L)]** Dissolve 5.0 g of boric acid in water and make 1 L in whole volume.

3.3 Apparatus The apparatus shall be in accordance with **5.3** of the text.

3.4 Sampling operation Carry out corresponding to the case of indophenol blue absorption spectrophotometry in **5.4** of the text.

3.5 Calculation of sampling volume of gas The calculation shall be in accordance with **5.5** of the text.

3.6 Preparation of sample solution for analysis After collecting the ammonia in flue gas into the absorption solution in **3.2 c**), operate corresponding to **6.1** of the text, and prepare.

4 Method of determination

4.1 Reagents and preparation of reagent solution

4.1.1 Reagents The reagents shall be as follows:

a) **Water**, class A2 specified in **JIS K 0557**.

b) **Sodium hydroxide**, specified in **JIS K 8576**.

c) **Ammonium chloride**, specified in **JIS K 8116**.

d) **Ammonium sulfate**, specified in **JIS K 8960**.

e) **Boric acid**, specified in **JIS K 8863**.

4.1.2 Preparation of reagent solution The preparation of reagent solution shall be as follows:

a) **Sodium hydroxide solution (100 g/L)** Dissolve 10 g of sodium hydroxide in water and make 100 ml in whole volume.

b) **Reference stock solution of ammonium ion (0.1 mg/ml)** That shall be in accordance with **7.1.2 b) 3)** of the text.

c) **Reference solution of ammonium ion I (0.01 mg/ml)** Take accurately 25.0 ml of the reference stock solution of ammonium ion (0.1 mg/ml) in **b)** into a 250 ml volumetric flask, and add water up to the marked line.

d) **Reference solution of ammonium ion II (0.001 mg/ml)** Take correctly 25.0 ml of the reference solution of ammonium ion I (0.01 mg/ml) in **c)** into a 250 ml volumetric flask, and add water up to the marked line.

e) **Reference solution of ammonium ion III (0.000 1 mg/ml)** Take accurately 25.0 ml of the reference solution of ammonium ion II (0.001 mg/ml) in **d)** into a 250 ml volumetric flask, and add water up to the marked line.

f) **Absorption solution**, prepared in **3.2 c**).

4.2 Apparatus The apparatus shall be as follows:

a) **Potentiometer** A high input resistance potentiometer of 1 mV (for example, ion meter, digital type pH-mV meter, pH-mV meter with scale-up span).

b) **Membrane type ammonia electrode**

c) **Magnetic stirrer** The stirrer giving no change of temperature of the solution by generated heat due to rotation.

4.3 Preparation of working curve The operation shall be carried out as follows:

a) Pour 100 ml of the reference solution of ammonium ion III (0.000 1 mg/ml) into a 200 ml Erlenmeyer flask, add 1 ml of sodium hydroxide solution and mix it as swinging gently(1).

b) Immediately, immerse the membrane type ammonia electrode(2), mix the solution with a magnetic stirrer at the constant speed so as not to change the temperature of solution. After the indication has been stabilized, measure the potential(3) (4) (5). At the same time, measure the temperature of the solution.

c) Carry out the operations of **a)** and **b)** for the reference solution of ammonium ion II (0.001 mg/ml) and the reference solution of ammonium ion I (0.01 mg/mol), and measure the potential(2) (3) (4) (5). In this measurement, adjust the temperature of the solution within ± 1 °C compared to the temperature measured in the operation **b)** for the reference solution III.

d) Take the concentration of ammonium ion (mg/ml) on the logarithmic axis of semi-logarithmic section paper, and prepare the relation curve(6) between the concentration of ammonium ion and the potential (mV).

Notes (1) At this time, pH value of this solution becomes approximately pH 12. At pH value of not less than 11, ammonium ion is converted to ammonia, care should be taken because ammonia volatilizes easily.

(2) For membrane type electrode, care should be taken because the membrane would be flawed if the membrane of the glass film surface of the inside glass electrode is pressed strongly.

(3) If the membrane of membrane type electrode stains, the potential becomes unstable, and the response speed gets slower.

(4) If the mixing with stirring is excessively strong, bubbles cover the membrane to cause errors.

(5) The response time of ammonia electrode is 3 min to 5 min for 0.000 1 mg/ml of ammonium ion concentration and is 2 min to 3 min for 0.001 mg/ml of ammonium ion concentration.

(6) The potential difference between the reference solution of ammonium ion of 0.000 1 mg/ml and that of 0.01 mg/ml falls in the range of 110 mV to 120 mV. The working curve for ammonium ion concentration up to near 0.000 1 mg/ml becomes a straight line.

Remarks : When the direct-reading instrument is employed, use the reference solution of ammonium ion III and the reference solution of ammonium ion I instead of preparation of the working curve and carry out several times to calibrate the scale (span adjustment) for each analysis repeatedly.

4.4 Operation for determination The operation shall be carried out as follows:

a) Take an adequate amount *v* ml of the sample solution for analysis prepared in **3.6** (containing 0.01 mg to 1 mg as NH_4^+) into a 200 ml beaker, and add sodium hydroxide solution little by little to make weak alkaline (approximately pH 7.5). Then, transfer it into a 100 ml volumetric flask with washing by water and add water up to the marked line. Transfer the solution into a 200 ml Erlenmeyer flask. The volumetric flask shall not be washed.

b) Adjust the solution temperature within ± 1 °C of the solution temperature at which the working curve was prepared in **4.3**.

c) Carry out the operations of **4.3 a**) and **4.3 b**), and obtain the concentration of ammonium ion (mg/ml) from the working curve prepared in **4.3 d**).

d) Take v ml of the absorption solution into a 200 ml beaker, operate corresponding to **a**) to **c**) and obtain the blank test value of ammonium ion.

Remarks : The standard method of addition may be used. The operation shall be as follows:

Take 100 ml of the sample solution for analysis into an Erlenmeyer flask and add 1 ml of sodium hydroxide solution, then carry out the operation corresponding to **4.3 b**) and read out the potential E_0. Next, add each 1 ml of the reference solution of ammonium ion having 100 times to 1 000 times of the approximate concentration (C mgNH_4^+/ml) obtained from the working curve and measure the generated potentials E_1, E_2 and E_3. Take the potential difference from E_0 as ΔE_1, ΔE_2 and ΔE_3. Obtain the potential gradient S (7) near the measuring concentration from the working curve, obtain the respective ammonium ion concentration a_1, a_2, a_3 according to the following formula, and calculate the concentration of ammonium ion concentration a (mg/L) in the sample solution for analysis.

$$a_1 = \frac{c}{101 \times 10^{\Delta E_1/S} - 100}$$

$$a_2 = \frac{2 \times c}{102 \times 10^{\Delta E_2/S} - 100}$$

$$a_3 = \frac{3 \times c}{103 \times 10^{\Delta E_3/S} - 100}$$

$$a = 1/3\,(a_1 + a_2 + a_3)$$

Note (7) Potential gradient means the potential change when the ammonium ion concentration has changed 10 times. Separately, carry out the same operation with 100 ml of absorption solution and obtain the concentration of ammonium ion, b (mg/ml).

4.5 Calculation Calculate the concentration of ammonia in the sample gas according to the following formula.

$$C_v = \frac{1.24 \times (a - b) \times 100/v \times 250}{V_s} \times 1\,000$$

$$C_w = \frac{0.944 \times (a - b) \times 100/v \times 250}{V_s} \times 1\,000$$

$$C_w = C_v \times 0.760$$

where, C_v : volume concentration of ammonia in sample gas (volppm)

C_w : mass concentration of ammonia in sample gas (mg/m^3)

a : concentration of ammonium ion obtained in **4.4 c)** (mg/L)

b : concentration of ammonium ion obtained in the blank test of **4.4 d)** (mg/ml)

v : aliquot volume of the sample solution for analysis (ml)

V_s : sampling volume of gas under the standard condition calculated according to **3.5** (L)
(V_{SD} is used for dry gas volume, and V_{SW} is used for wet gas volume)

1.24 : volume of ammonia (NH_3) equivalent to 1 mg of NH_4^+ (ml) (the standard condition)

0.994 : mass of ammonia (NH_3) equivalent to 1 mg of NH_4^+ (mg)

0.760 : mass concentration of ammonia (NH_3) equivalent to 1 volppm of ammonia (mg/m^3)

Annex 2 (normative)

Analytical method for the case of interference by coexisting sulfur oxides, etc. in the indophenol blue absorption spectrophotometry

1 Scope This Annex specifies the analytical method of ammonia in the case of interference by coexisting sulfur oxides, etc. at the measuring time of ammonia in flue gas by indophenol blue absorption spectrophotometry.

2 Outline of analytical method The outline of analytical method shall be in accordance with Annex 2 Table 1.

Annex 2 Table 1 Outline of analytical method

Outline of analytical method		
Summary	Sampling method	Range of determination volppm (mg/m^3)
After absorbing ammonia in flue gas into hydrogen peroxide aqueous solution, expel ammonia by adding sodium hydroxide solution in an expelling apparatus for ammonia, collect it in boric acid solution, then add sodium phenol-pentacyanonitrosylferrate (III) solution and sodium hypochlorite solution. Indophenol blue is generated, measure its absorbance (640 nm).	Absorption bottle method Absorption solution: hydrogen peroxide aqueous solution (1+9) Absorption volume: 50 ml × 2 bottles Standard sampling volume: 20 L	1.6 to 15.5 (1.2 to 11.8)

3 Sampling method of gas

3.1 Sampling of gas The sampling of gas shall be in accordance with clause **5** of the text.

3.2 Reagents and preparation of absorption solution The reagents and preparation of absorption solution shall be as follows:

a) **Water**, class A2 specified in **JIS K 0557**.

b) **Hydrogen peroxide**, specified in **JIS K 8230**.

c) **Sodium hydroxide**, specified in **JIS K 8576**.

d) **Absorption solution**, hydrogen peroxide aqueous solution (1+9).

e) **Sodium hydroxide solution (8 mol/L)** Dissolve 32 g of sodium hydroxide in water to make 100 ml.

f) **Boric acid solution (5 g/L)** Dissolve 5.0 g of boric acid in water to make 1 L in whole volume.

3.3 Apparatus The apparatus shall be in accordance with **5.3** of the text.

3.4 Operation of sampling The operation of sampling shall be in accordance with **5.4** of the text.

3.5 Calculation of sampling volume of gas The calculation shall be in accordance with **5.5** of the text.

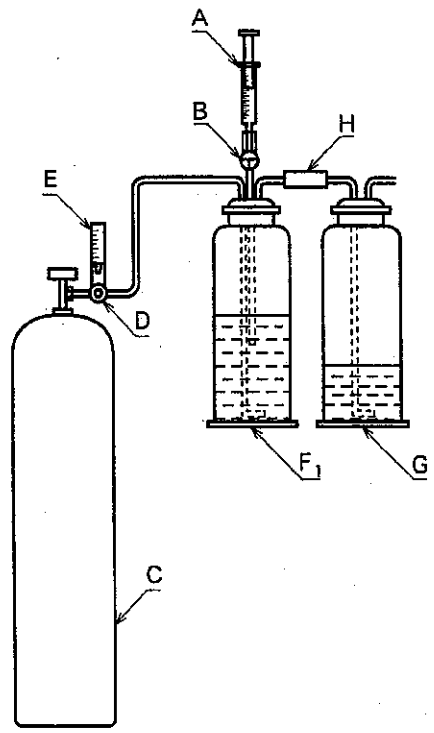
3.6 Expelling operation of ammonia The expelling operation of ammonia shall be as follows:

a) After taking off the absorption bottles (F_1, F_2) and transferring the solution contained in the absorption bottle (F_2) into the absorption bottle (F_1) by washing with water, connect the absorption bottle (F_1) to the ammonia expelling apparatus in Annex 2 Fig. 1.

b) Fix a syringe of plastics (A) in which 5 ml of sodium hydroxide solution (8 mol/L) are sucked.

c) Introduce the sodium hydroxide solution contained in the syringe (A) into the absorption bottle (F_1) and make it alkaline (at least pH 13).

d) Open the valve of nitrogen cylinder, adjust the flow rate to approximately 2 L/min and absorb the generated ammonia into the boric acid solution in the absorption bottle (G). The expelling time shall be 100 min.



A : Syringe of plastics [sodium hydroxide solution (8 mol/L), 5 ml]

B : Two way stop cock of plastics

C : Nitrogen cylinder

D : Control valve of flow rate

E : Flowmeter

F_1 : Absorption bottle (absorption solution after absorbing sample gas + washings)

G : Ammonia absorption bottle [boric acid (5 g/L), 50 ml]

H : Silicone rubber tube

Annex 2 Fig. 1 An example of ammonia expelling apparatus

3.7 Preparation of sample solution for analysis The preparation of sample solution for analysis shall be as follows:

a) After carrying out the operation of **3.6**, transfer the solution contained in the absorption bottles (F_1, F_2) into a 300 ml beaker and further wash the absorption bottles, etc. with absorption solution [boric acid solution (5 g/L)].

b) Transfer the solution contained in the beaker, with washing by absorption solution, into a 250 ml volumetric flask and further add absorption solution up to the marked line. Take this solution as the sample solution for analysis.

4 Method of determination The method of determination shall be in accordance with **7.1** of the text.

Related standard:

JIS K 0804 *Gas detector tube measurement system (Length-of-stain type)*

Errata for JIS (English edition) are printed in *Standardization Journal*, published monthly by the Japanese Standards Association, and also provided to subscribers of JIS (English edition) in *Monthly Information*.

Errata will be provided upon request, please contact:
Standardization Promotion Department, Japanese Standards Association
4-1-24, Akasaka, Minato-ku, Tokyo, 107-8440 JAPAN
TEL. 03-3583-8002 FAX. 03-3583-0462

100% Recycled paper